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Rolls-Royce	ENGINEERING SPECIFICATION	RRES 90005
		Revision D
		First Issued 10 Feb 2009
		Revised 26 Nov 2019

Traceability of Serialised Items

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Jurisdiction	Export Control Classification (Rating)	Date
U.S.	No Export Control Data	1 Oct 2019
U.K.	Not Listed	1 Oct 2019
GER	NLR	1 Oct 2019

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Retention Category A

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1 SCOPE:

1.1 This specification defines the requirements for traceability of serialised items.

2 REASON FOR USE:

2.1 This specification describes the data retention requirements to ensure an adequate record of manufacturing processing.

2.2 The specification shall apply in its entirety when it is called out on a Component Definition.

3 APPLICABLE DOCUMENTS:

The following publications form a part of this document to the extent specified herein. The applicable issue shall be the current issue, unless a specific document issue is specified. When the referenced document has been cancelled and no superseding document has been specified, the last published issue of that document shall apply. Additionally, contact your sector's Rolls-Royce Technical Authority to obtain guidance/approval on action to remove or replace the cancelled reference by a drawing and/or specification change. In the event of a conflict between the text of this document and the references cited herein, the text of this document takes precedence.

3.1 Rolls-Royce Documents: Available internally from the A to Z Infocentre website http://www.infocentre.rolls-royce.com/technical_library and Teamcenter. Available externally from the Global Supplier Portal: https://suppliers.rolls-royce.com/".

RRES 90000 Engineering Control of Manufacturing and Repair Source and Method

4 DEFINITIONS:

Billet:
A rolled, extruded or forged product intended for further hot working.

Condition of Supply:
Items delivered to Rolls-Royce that require further machining, manufacturing, and/or processing steps before the item is ready for assembly. Examples include: bar stock, sheet stock, plate stock, rolled rings, forgings, square cut forgings, and castings. Also commonly referred to as Raw Material.

Conversion:
Use of a mechanical or thermomechanical process to convert master heat material or master powder blends into intermediate forms intended for subsequent forging; examples are hot isostatic pressing, extrusion, swaging, and rolling.

Fixed Process Document:
A document required and controlled by RRES 90000 which defines manufacturing information such as the method of manufacture, equipment, operation sequence, methods, parameters, and control techniques.

Ingot:
A casting (usually from a master heat) intended for working or remelting.

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Item:
A non specific term used to denote any part, unit, assembly, or product.

Lot:
All items produced in one time period by a single source, maintained, sampled, tested and inspected as a common unit and, when applicable, of the same component size. Also commonly referred to as Batch.

Master Heat:
A quantity of alloy processed as a unit through melting and refining and subject to subsequent qualification by compositional analysis and additional testing as required by the applicable material specification.

May:
Denotes a permissible practice or action.

Melt Charge:
The material stock (raw material, titanium sponge, revert, etc.) used in the initial melting process.

Mill Certification:
A document that certifies that a particular lot of material meets a specific set of requirements; includes the mill source, the master heat identification number, the production date, and the specification(s) to which the material complies.

Mult:
A piece of stock, cut from bar or billet, from which a forging is made; also known as a "multiple" or "slug".

Must:
Is used only for legislative or regulatory requirements (e.g. Health and Safety) and shall be complied with.

Non-Drawn Part:
A sub-assembly component that does not have its own part number, but instead whose engineering definition is entirely contained within the Component Definition of the parent assembly.

Serial Number:
A formatted string of data (comprising of upper case letters, numbers, and or hyphen(-)) that provides a means of identifying a specific individual item. The hyphen shall not be used in the first or last position nor shall it be used consecutively. The serial number is prefixed by the TEI "SER ". Max length is 15 characters.

Shall:
Denotes a mandatory requirement. Departure from such a requirement is not permissible without formal agreement with the design authority and shall be properly documented.

Should
Denotes a recommendation or advice and is expected to be followed unless good reasons are stated for not doing so.

Source:
Any organization that is responsible for supplying items to Rolls-Royce in response to a Rolls-Royce purchase order or a Rolls-Royce production order.

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Special Processes:
Processes that change the material in some way (physical, metallurgical, etc.) that cannot be easily or reliably evaluated without using destructive means . Examples of such processes include: heat treatment, non-conventional machining, etching, and plating.

Traceability:
The ability to retrieve information about an event or events that have occurred during the manufacture of a particular item.

5 GENERAL REQUIREMENTS:

5.1 Traceability Information Requirements:

5.1.1 The source shall establish a system that generates and maintains traceability information in accordance with Table 1.

5.1.2 The traceability information requirements for a particular item are identified by the “dash number” (-1 through -6) associated with this specification (e.g., RRES 90005-1 as specified on the Component Definition).

5.2 Serial Number Requirements:

5.2.1 Each item shall be identified with a serial number in accordance with the marking requirements stated on the Component Definition.

5.2.2 Serial Number Format

5.2.2.1 Each serial number consists of two elements: 1) a Source Code and 2) an Item Identifier. Both elements are required to make an acceptable serial number. See Figure 1. Acceptable characters within serial numbers include: Capital letters A thru Z, numerals 0 through 9, and the special character “-“ (dash – ASCII hexadecimal code x2D). No blanks are allowed.

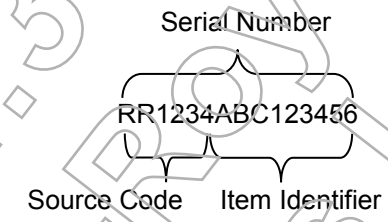

FIGURE 1 - SERIAL NUMBER FORMAT

5.2.2.2 Externally Manufactured Items:

5.2.2.2.1 Source Codes for items manufactured by external suppliers are assigned by Rolls-Royce as the SAP vendor code (SVC) and consists of up to 6 characters. Item Identifiers are assigned by the source and shall not exceed 9 characters. For example, 202309ABC112233.

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5.2.2.3 Internally Manufactured Items:

5.2.2.3.1 Source Codes for items manufactured by Rolls-Royce consist of up to 6 characters with the first two characters being “RR” followed by the 4 character Rolls-Royce SAP plant code, or other approved plant-specific code (generally, codes that are already established; non-SAP codes are not recommended for new manufacturing locations). Item Identifiers are assigned by Rolls-Royce and shall not exceed 9 characters. For example, RRINCH654321ABC.

5.2.3 The serial number format specified in Section 5.2.2 is optional for items considered Condition of Supply.

5.2.4 Non-drawn parts do not require serialization, but have the same requirement for traceable records as the parent assembly unless otherwise indicated on the Component Definition. Traceable records for non-drawn parts shall be stored with or linked to the records for the parent assembly.

5.2.5 Serial Number Uniqueness:

5.2.5.1 The source, whether external or internal, shall ensure that each serial number is used once and never duplicated on any Rolls-Royce item. Serial numbers shall not be duplicated within a Source Code regardless of part number.

5.2.6 Serial Number Reuse:

5.2.6.1 Serial numbers shall not be reused from scrapped components.

6 QUALITY ASSURANCE:

6.1 Responsibility for Inspection:

The source shall inspect each item to verify that proper method, location, and serial number has been marked on each item in accordance with the marking requirements stated on the Component Definition.

6.2 Retention of Records:

All records required to assure product conformance to this specification shall be maintained by the source as required by contract or government regulation.

6.3 Preparation for Delivery to Rolls-Royce:

Each shipment of items shall include a listing of serial numbers in the shipment.

7 ACKNOWLEDGEMENT:

Sources shall refer to this specification in all quotations and when acknowledging purchase orders.

8 REJECTIONS:

Any item which fails to meet the requirements of this specification shall be subject to rejection.

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TABLE 1 – MINIMUM TRACEABILITY REQUIREMENTS

	RRES 90005 Dash Number - Level of Traceability					
Traceability Information	-1	-2	-3	-4	-5	-6
Identification of melt charge inputs	Required					
Mill certification and material specification	Required	Required	Required			
Forgings* Only: Data on processing used for conversion to intermediate forms prior to forging: • Conversion source(s), method(s), and parameters	Required	Required (except AMS materials)				
Forgings* Only: Full ingot position, including**: • Ingot identification and location • Billet identification and location • Mult identification and location	Required					
Forging, Casting, and/or Heat Treatment process data, including: • Source of processing • Tooling and equipment • Time and date • Run identification number • Methods and parameters (including atmosphere and cooling rate, if controlled) • Material preheat temperature • Certification records for operator(s) and equipment	Required	Required	Required			
Furnace position / loading pattern	Required	Required (Loading Pattern Only)				
All data reports required by all applicable specifications	Required	Required	Required	Required		
Forging/casting lot identification	Required	Required	Required	Required		
Data for each Special Process, including: • Source of processing • Tooling and equipment • Time and date • Run identification number • Methods and parameters (includes atmosphere and cooling rate, if controlled) • Material preheat temperature • Certification records for operator(s) and equipment	Required	Required	Required	Required		
Final inspection and test data (including non-destructive evaluation) identified on the Component Definition	Required	Required	Required	Required	Required	
List of all subassembly part and serial numbers, if applicable	Required	Required	Required	Required	Required	
Designation and revision level of all process work instructions for each special process	Required	Required	Required	Required	Required	
Designation and revision level of each Fixed Process Document, if applicable	Required	Required	Required	Required	Required	
Certificate of Conformance	Required	Required	Required	Required	Required	
Serialisation only - No traceability information required						Required

* Includes parts made from non-drawn or supplier-controlled forgings

** Billet Location not required for billet sizes <105 mm diameter. Other items are required unless otherwise agreed with the Rolls-Royce Technical Authority.

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9 NOTES:

9.1 Revision bars indicate changes from the previous issue of this specification and are located in the left margin.

Issue	Change (ESC) Number	Change Summary (In Descending Order)	Date
D	ESC500037975969	Condition of supply serialisation requirements clarified, RFS1938.	26 Nov 2019
C	ESC500036250212	RFS 1014, 1359, 1818: - Condition of supply definition clarified. RFS 1359 - Section 4 – added definition of Non-drawn part, standard RRES definitions and clarified the condition of supply definition. RFS 1818 - Para. 5.2.2.3.1 – allow the use of other plant specific codes to permit current manufacturers to proceed as usual. FRS 1014 - Add Para. 5.2.4 –Clarified the serialization requirements for non-drawn parts. RFS 1818.	25 Jun 2019
B	ESC500023027376	- Max length of 15 characters added to serial number definition. - Table 1 asterisk note: resolving conflict with RRMS 30000 - RFS 998, 1024	17 Feb 2015
A	ESC500013712007	- Revised Scope and Definitions sections - Section 5 extensively revised to give better clarification of serial number requirements - Table 1 simplified	20 Oct 2010
Initial	---	This global specification is needed for the XWB engine program. EDNS04000002605	10 Feb 2009

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